



Thermofluid Heating Pipe System

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1.Abstract

This project presents the modeling and simulation of a domestic hot water system, focusing on the heating process within a pipe through which water flows. A mathematical model capturing the heat transfer, fluid properties, and heat loss to the environment was developed. MATLAB's ODE solver was used to simulate transient temperature changes and system efficiency under varying conditions, including heater power and insulation presence. Results demonstrate the impact of insulation and flow rate on heating efficiency and outlet temperature, providing insights into system optimization.

2. Introduction

Domestic hot water systems are essential in residential and commercial settings for providing hot water for various uses. Efficient operation requires understanding the thermal behavior of the fluid inside piping systems, the influence of heating elements, flow conditions, and heat loss mechanisms. This study develops a mathematical model of such a system and simulates its behavior to assess performance and efficiency.

3. Device Design

The system consists of a heated pipe section where water flows and is heated by an electric heater. The pipe is exposed to ambient conditions, causing heat losses. Key design parameters include pipe diameter and length, fluid properties, flow rate, and heater power. Insulation can be applied to reduce heat losses.

4. Modeling

The energy balance on the fluid in the pipe is modeled by considering heat input from the heater, convective heat losses to the environment, and heat carried away by the flowing fluid. The model assumes:

- Fluid properties depend on the fluid type (e.g., water, oil, alcohol, mercury).
- Heat loss to the environment is modeled by the equation :

$$q_{loss} = h_{loss} \cdot A \cdot (T - T_{env})$$

q_{loss} : is the rate of heat loss to the environment (W),

h_{loss} : is the heat loss coefficient (W/m²·K),

A : is the surface area of the pipe (m²),

T : is the fluid temperature inside the pipe (°C),

T_{env} : is the ambient temperature (°C).

$$\frac{dT}{dt} = \frac{Q - q_{\text{loss}} - \dot{m} \cdot c_p \cdot (T - T_{\text{in}})}{m \cdot c_p}$$

dT/dt is the rate of change of temperature with time (°C/s),

Q is the heater power (W),

$\dot{m}$ is the mass flow rate of the fluid (kg/s),

c_p is the specific heat of the fluid (J/kg·K),

T_{in} is the inlet temperature (°C),

m is the mass of the fluid in the pipe (kg).

```

function [t, T, efficiency] = simulateHeating(pipe_diameter, pipe_length, ...
    fluid_type, heater_power, flow_rate, T_in, h_loss)

% Physical properties based on fluid type
switch fluid_type
    case 'Water'
        cp = 4186; density = 1000;
    case 'Oil'
        cp = 2000; density = 900;
    case 'Alcohol'
        cp = 2500; density = 790;
    case 'Mercury'
        cp = 140; density = 13500;
    otherwise
        cp = 4186; density = 1000;
end

% Geometric and physical calculations
cross_area = pi * (pipe_diameter / 2)^2;
volume = cross_area * pipe_length;
mass = density * volume;

% Environment parameters (adjust if needed)
T_env = 25;
surface_area = pi * pipe_diameter * pipe_length;

% Time span
tspan = [0 500]; % seconds

% ODE Solver
[t, T] = ode45(@(t, T) heatingODE(t, T, T_in, flow_rate, heater_power, ...
    mass, cp, h_loss, surface_area, T_env), tspan, T_in);

% Efficiency calculation
heat_used = flow_rate * cp * (T - T_in);

if heater_power <= 0
    efficiency = zeros(size(T));
else
    efficiency = (heat_used / heater_power) * 100;
end

end

```

```

function dTdt = heatingODE(t, T, T_in, flow_rate, Q, mass, cp, ...
                           h_loss, surface_area, T_env)

if mass < 1e-6
    dTdt = 0;
else

% Heat loss term
q_loss = h_loss * surface_area * (T - T_env);

% Energy balance
dTdt = (Q - q_loss - flow_rate * cp * (T - T_in)) / (mass * cp);

end
end

```

5. Simulation

The model is implemented in MATLAB using the ode45 solver for time integration. Inputs include pipe dimensions, fluid type, heater power, flow rate, inlet temperature, and heat loss coefficient (variable depending on insulation). The simulation runs for 500 seconds to capture transient and steady-state behavior. Efficiency is computed as the ratio of heat gained by fluid to heater power input.

% Button pushed function: RunButton

function RunSimulationPushButton(app, event)

```
pipe_diameter = app.PipeDiametermEditField.Value;
pipe_length = app.PipeLengthmEditField.Value;
fluid_type = app.FluidTypeDropDown.Value;
flow_rate = app.FlowRatekgsKnob.Value;
T_in = app.InletTempCEditField.Value;
heaterOn = app.HeaterSwitch.Value;
insulationOn = app.InsulationSwitch.Value;

if strcmp(heaterOn, 'On')
    app.StatusIndicatorLamp.Color = [0 1 0];
    heater_power = app.HeaterPowerWKnob.Value;
else
    app.StatusIndicatorLamp.Color = [1 0 0];
    heater_power = 0;
end

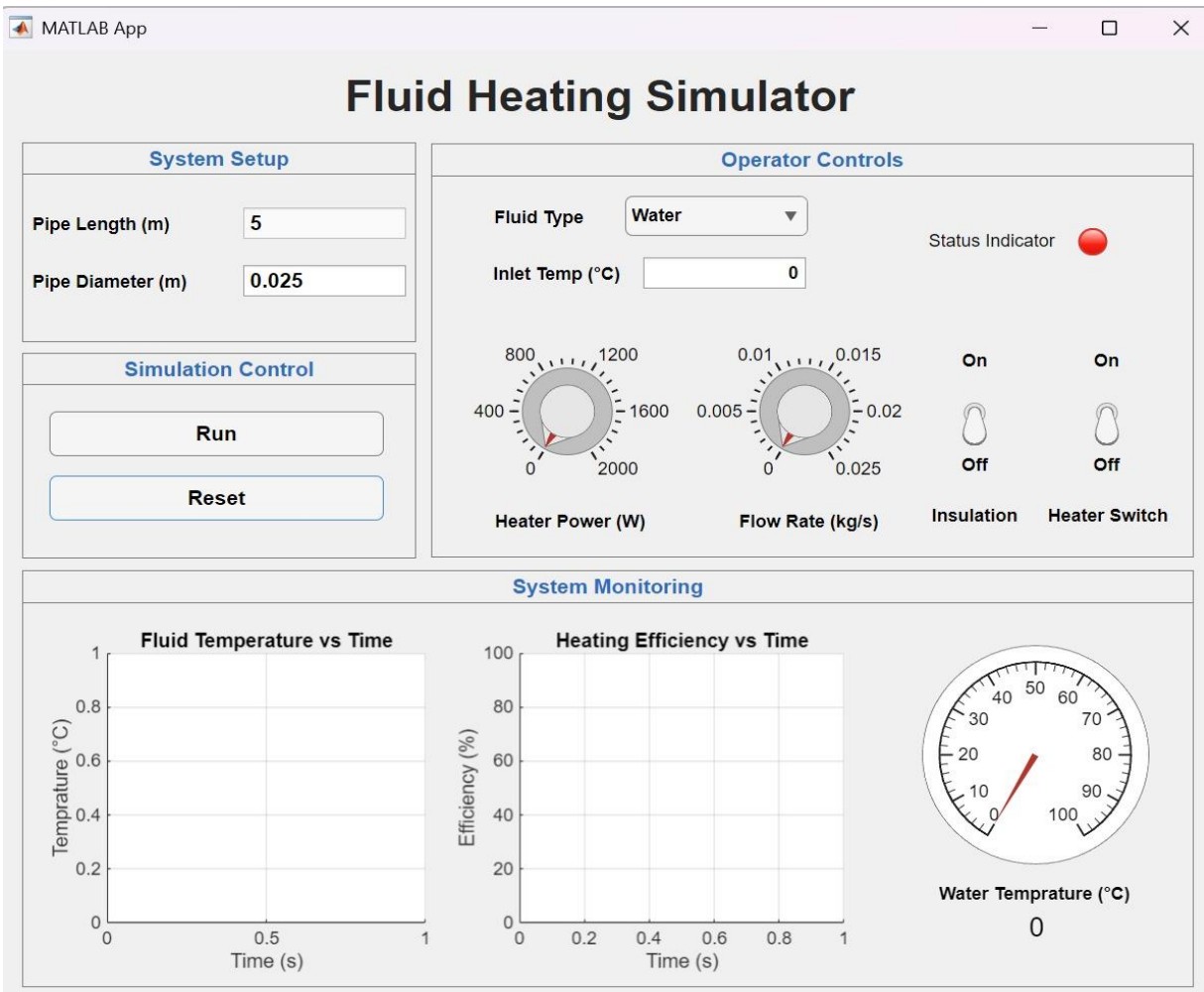
    if strcmp(insulationOn, 'On')
        h_loss = 5; %W/m2·K
    else
        h_loss = 25;
    end

[t, T, efficiency] = simulateHeating(pipe_diameter, pipe_length,...
    fluid_type, heater_power, flow_rate, T_in, h_loss);

% Plot temperature
plot(app.UIAxes, t, T, 'b-', 'LineWidth', 2);
grid(app.UIAxes, 'on');

% Plot efficiency
plot(app.UIAxes2, t, efficiency, 'r--', 'LineWidth', 2);
ylim(app.UIAxes2, [0 100]);
grid(app.UIAxes2, 'on');

app.WaterTempratureCGauge.Value = T(end);
app.Label.Text = num2str(T(end), '%.1f');
```



6. Operation

A MATLAB App was developed to allow interactive control of system parameters, including toggling the heater and insulation on/off, adjusting flow rate, and viewing temperature and efficiency plots. The system updates simulation results dynamically based on user inputs.

7. Simulation results confirm expected trends:

*** With heater ON and insulation applied (lower heat loss coefficient), the fluid temperature rises faster and reaches higher steady-state values.**

*** Without insulation (higher heat loss), the temperature rise is slower and lower steady-state temperature is achieved.**

*** Efficiency plots show corresponding improvements with insulation.**

*** Flow rate affects temperature rise rate and outlet temperature due to thermal mass and residence time.**

These results align qualitatively with real-world expectations and literature, validating the model.

9. Conclusion

This project successfully developed and simulated a mathematical model of a domestic hot water system, incorporating heat transfer, fluid flow, and environmental heat loss. The model and MATLAB App provide a useful tool for exploring system behavior and optimizing design parameters such as insulation and flow rate for energy efficiency.